

1) Graphplan

W9?

2) AO*

3) CSP

W11

4) Line-drawing

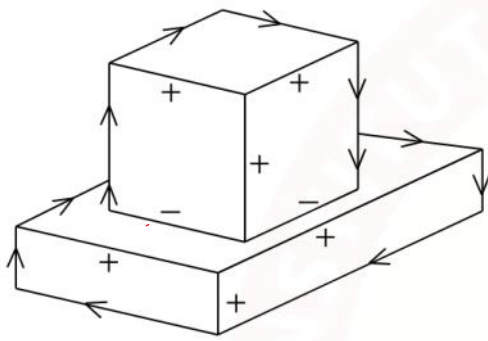
5) Arc-Consistency

AC-1

AC-3

6) i-consistency

7) Model-Based Diagnosis



Convex & Concave Edge

- + : Convex Edge
- : Concave Edge
- < : Plane to the right

Vertex Categories - 1) $\vee$ joint

2) W joint

3) T joint

4) L joint

Implication - $A \rightarrow B$

$$A \supset B$$

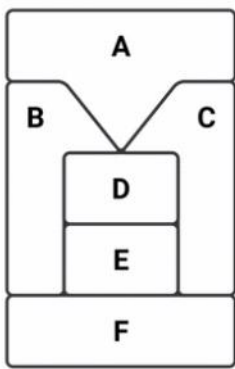
A	B	$A \rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

CSP is defined as a triple $\langle X, D, C \rangle$

X - Variables

D - Domains

C - Constraints / Relations



$$D_A = \{r, b\}$$

$$D_B = \{r, b, g\}$$

$$D_C = \{r, b, g\}$$

$$D_D = \{r, g\}$$

$$D_E = \{b, g\}$$

$$D_F = \{r, b\}$$

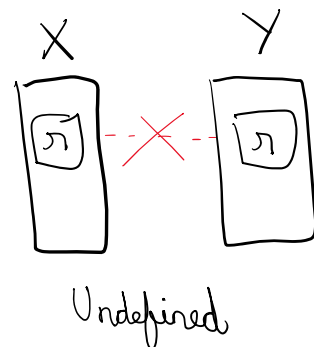
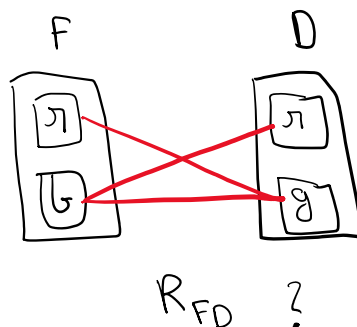
$$X = \{A, B, \dots, F\}$$

$$D = \{D_A, D_B, \dots, D_F\}$$

$$C = \{R_{AB}, R_{AC}, R_{AD}, R_{BA}, R_{BC}, R_{BD}, R_{BE}, R_{BF}, R_C, \dots, R_D, \dots, R_E, \dots, R_F, \dots\}$$

$$R_{AB} = (D_A \times D_B) / \{\langle x, y \rangle, x = y\}$$

Q1) R_{FD} is not undefined

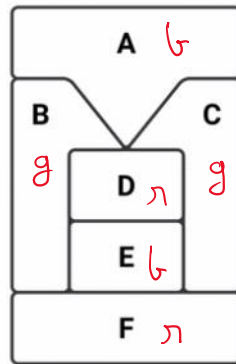


Q3) $D_A = \{r, b\}$ $D_F = \{r, b\}$

Q3) $D_A = \{r, b\}$ $D_F = \{r, b\}$

$R_{AF} = \{ \langle r, b \rangle, \langle b, r \rangle \}$

Q5) ABCDEF



$D_A = \{r, b\}$

$D_B = \{r, b, g\}$

$D_C = \{r, b, g\}$

$D_D = \{r, g\}$

$D_E = \{b, g\}$

$D_F = \{r, b\}$

CSP - 2

Q1)

$D_X = D_Y = D_Z = \{a, b, c, d\}$

$R_{XY} = R_{YZ} = \{ (a,b), (a,c), (a,d), (c,b), (d,b), (d,c) \}$

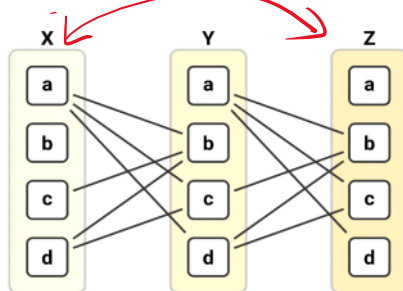
3 relations

symm.

reflexive

transitive

1) What is true about the diagram shown below.



$R_{XZ} = \{ (a,b), (a,c), (d,b) \}$

Arc-Consistency - (X, Y) are arc consistent iff

$\forall a \in D_X, b \in D_Y \quad \langle a, b \rangle \in R_{XY}$

$\langle a, b \rangle \in R_{XY} \rightarrow a \in D_X \text{ and } b \in D_Y$

$X = \{r, b\} \quad Y = \{b, g\}$

X & Y are consistent $\hookrightarrow$

$$\{ \langle n, b \rangle, \langle b, b \rangle, \langle n, g \rangle, \langle b, g \rangle \} \in R_{xy}$$

$$(x \times y) \in R_{xy}$$

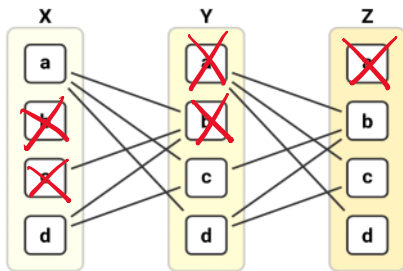
Revise $((X), Y)$ - If $n \in D_x$ and $\overset{\text{not}}{\uparrow} y \in D_y$ st $\langle x, y \rangle \in R_{xy}$
 then remove n from D_x

$$D_x = D_x / \{n\}$$

$$D_X = D_Y = D_Z = \{a, b, c, d\}$$

$$R_{XY} = R_{YZ} = \{ (a,b), (a,c), (a,d), (c,b), (d,b), (d,c) \}$$

1) What is true about the diagram shown below.



Revise $((X), Y)$

$$D_x = \{a, \cancel{b}, c, d\}$$

$$= \{a, \cancel{c}, d\} = \underline{\{a, d\}}$$

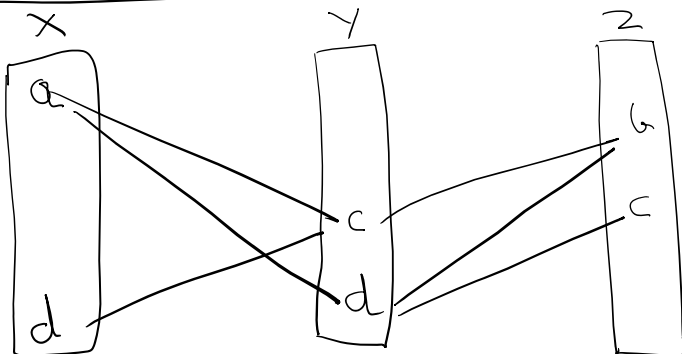
$$D_y = \{\cancel{a}, \cancel{b}, c, d\}$$

$$= \{\cancel{b}, c, d\}$$

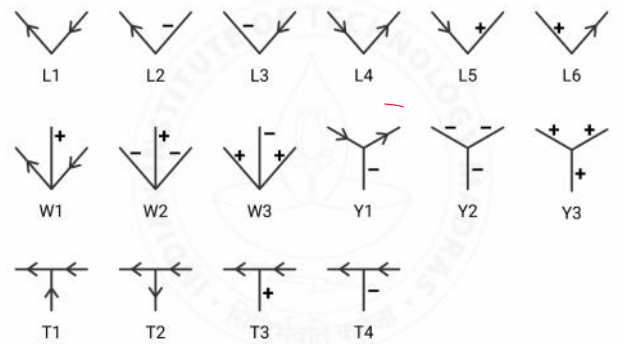
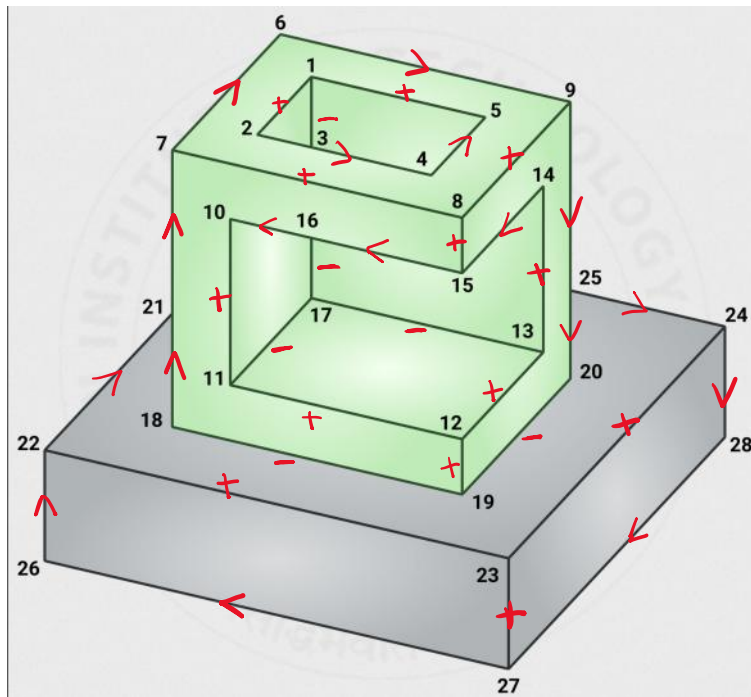
$$= \underline{\{c, d\}}$$

$$D_z = \{\cancel{a}, \cancel{b}, c, d\}$$

$$= \underline{\{b, c, d\}}$$



Pencil / Eraser



$$V_1 - W_3$$

$$V_3 - W_1$$

$$V_{19} - W_2$$

$$V_2 - L_6$$

$$V_4 - L_4$$

$$V_6 - L_1$$

$$V_{17} - Y_2$$

$$V_{23} - Y_3$$

$$V_{16} - T_4$$

$$V_{21} - T_1$$

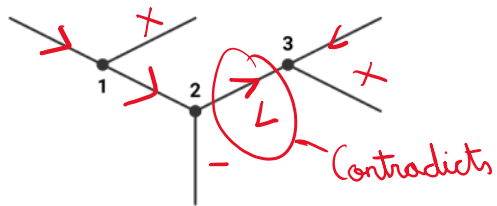
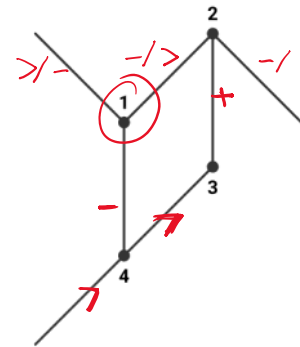
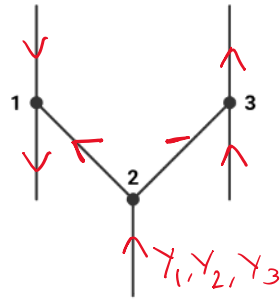
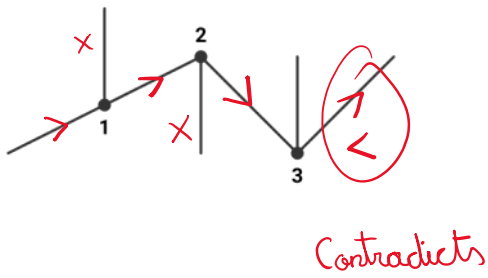
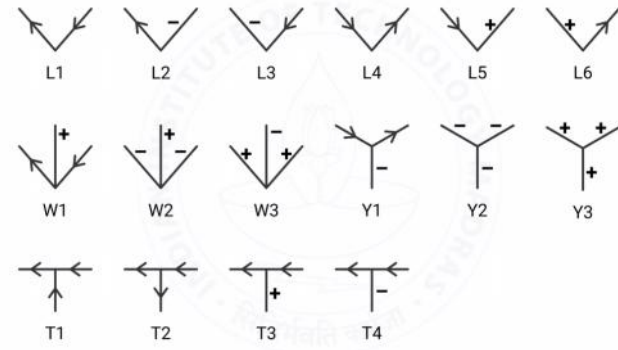
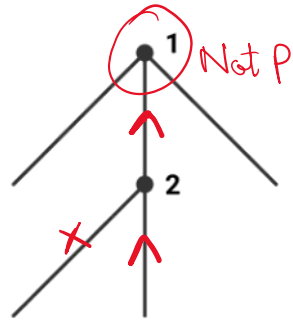
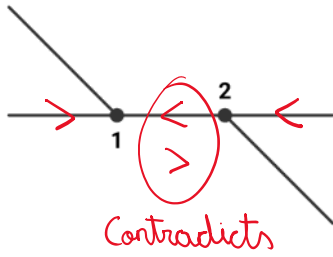
$$V_{25} - T_2$$

> - Material to the right

+ : Material inside right-angle

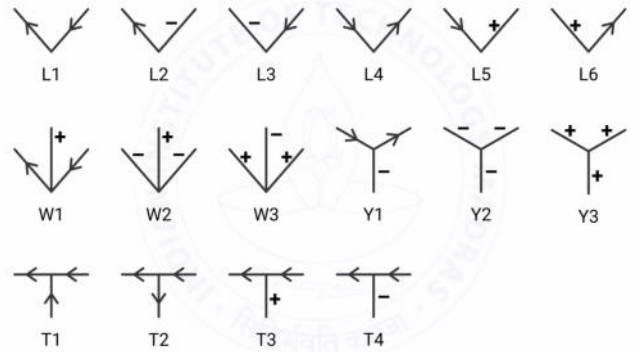
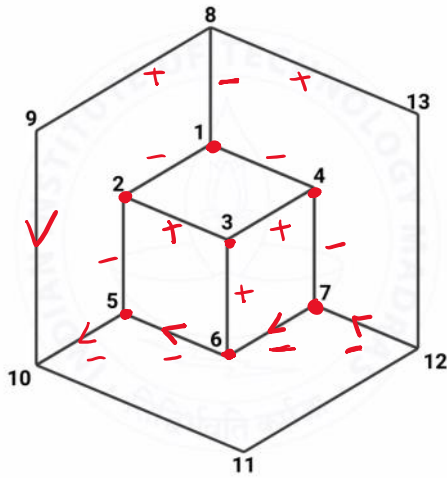
- : Material outside right angle

Don't use brain to infer faces & angles you can't see.

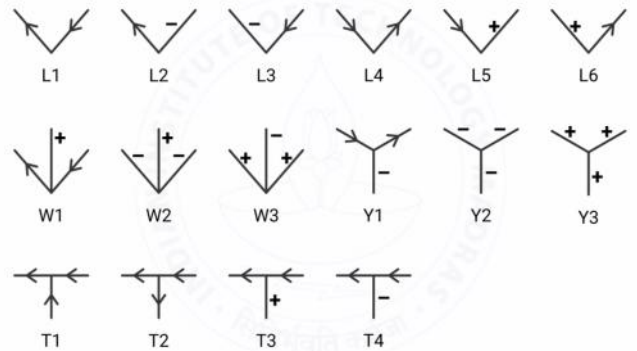
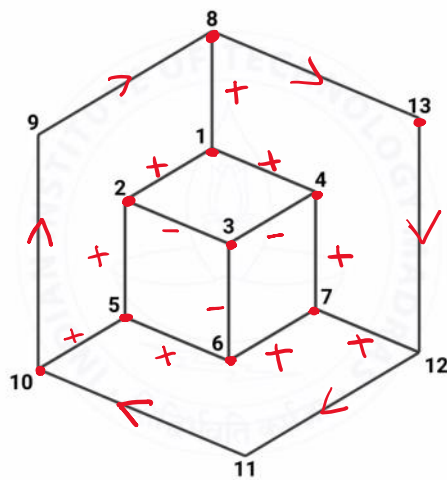


GA - 11

1



7

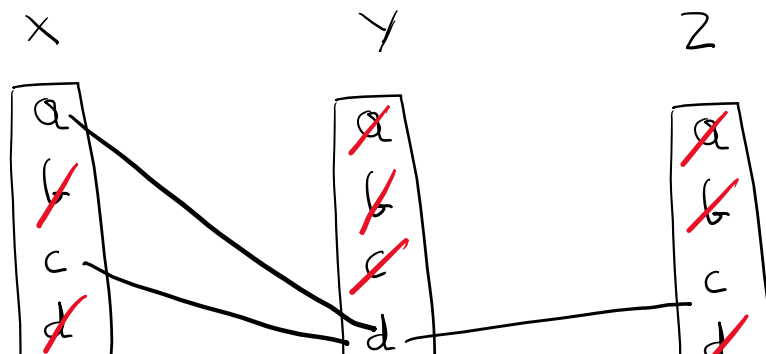


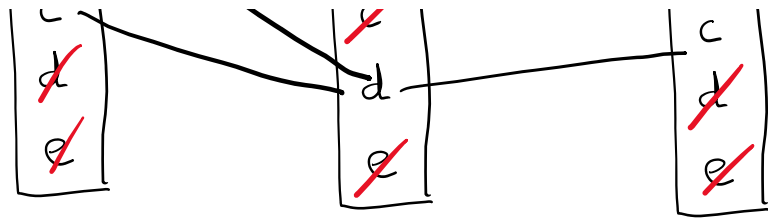
8

$$D_X = D_Y = D_Z = \{ a, b, c, d, e \}$$

$$R_{XY} = \{ (a,d), (b,c), (c,d), (e,c) \}$$

$$R_{YZ} = \{ (a,c), (a,b), (b,a), (b,c), (d,c), (e,b) \}$$





$a \in D_x \wedge \nexists b \in D_y \text{ s.t. } \langle a, b \rangle \in R_{x,y} \rightarrow \text{not consistent}$